AWS VPC Peering Project

Secure communication between isolated Virtual Private Clouds

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AWS Region: Asia Pacific (Mumbai) — ap-south-1
Project status: Infrastructure deployed; connectivity validation and laptop SSH troubleshooting partially complete

Project report based on the resources and test outputs recorded during implementation. Results are reported as observed; unsuccessful tests are not represented as successful.

1. Project Objective

The objective is to build two separate Amazon VPCs and establish private network communication between them using VPC peering. A public bastion host in VPC A provides an administrative entry point to private EC2 instances. Route tables and security groups control traffic between the VPCs and the instances.

2. Architecture Overview

Component	Configuration
AWS Region	ap-south-1 (Mumbai)
VPC A	ewb-vpc-a — 10.10.0.0/16
VPC A public subnet	ewb-a-public — 10.10.1.0/24; AZ ap-south-1a
VPC A private subnet	ewb-a-private — 10.10.2.0/24
VPC B	ewb-vpc-b — 10.20.0.0/16
VPC B private subnet	ewb-b-private — 10.20.1.0/24; AZ ap-south-1a
Bastion EC2	ewb-bastion — 10.10.1.10; public IPv4 3.111.147.64
App-A EC2	ewb-app-a — 10.10.2.10; private subnet in VPC A
App-B EC2	ewb-app-b — 10.20.1.10; private subnet in VPC B
VPC peering	ewb-a-b-peer — pcx-0eb7fcd4b731fddb4 (reported Active)

Logical traffic path: Laptop → public IPv4 of bastion in VPC A → App-A (10.10.2.10) or App-B (10.20.1.10). Traffic between the VPC CIDR ranges is routed through the active VPC peering connection.

3. Routing and Security Configuration

Route table	Required routes
ewb-rt-a-public	10.10.0.0/16 → local; 0.0.0.0/0 → ewb-igw-a; 10.20.0.0/16 → VPC peering
ewb-rt-a-private	10.10.0.0/16 → local; 10.20.0.0/16 → VPC peering
ewb-rt-b-private	10.20.0.0/16 → local; 10.10.0.0/16 → VPC peering

Security groups were configured for SSH (TCP 22) and ICMP tests. The bastion security group originally allowed SSH from the user's public IP (152.57.238.192/32). An inbound SSH rule was then added for the AWS-managed EC2 Instance Connect prefix list in ap-south-1, preserving the original rule. App-A and App-B were configured to allow SSH from the bastion private IP (10.10.1.10/32).

4. Implementation and Validation

Test	Observed result	Interpretation
Bastion identity	hostname returned ip-10-10-1-10.ap-south-1.compute.internal, confirmed through AWS EC2 Instance Connect.	Bastion shell access verified through EC2 Instance Connect.
Bastion → App-A TCP 22	Port 22 reachable	TCP connection from bastion to App-A's SSH port succeeded.
Bastion → App-B TCP 22	Port 22 reachable	TCP connection from bastion to App-B's SSH port succeeded across the peere
Bastion → App-A ICMP	4 packets transmitted, 0 received; 100% packet loss	Ping replies were not received. This does not by itself establish that TCP conne
Bastion → App-B ICMP	4 packets transmitted, 0 received; 100% packet loss	Ping replies were not received. ICMP/security-group rules and return path requi
Windows laptop → bastion TCP 22	TcpTestSucceeded: False; connection timed out	Direct laptop SSH remained unresolved at the time of reporting.

Test	Observed result	Interpretation
Windows SSH ProxyJump to App-A	Timed out connecting to bastion	ProxyJump configuration was recognized, but the laptop-to-bastion TCP timeou
SSH from bastion to App-A	Permission denied (publickey, ...)	Network path responded, but the bastion did not have a private key for App-A a
EC2 Instance Connect	Browser terminal opened after adding EIC path to bastion SG	Bastion shell access through browser-based EC2 Instance Connect verified.

5. Outstanding Work

• Investigate why laptop-originated TCP connections to the bastion public IP on port 22 time out, despite successful browser-based EC2 Instance Connect.

• Complete an end-to-end SSH ProxyJump test from the laptop using the local private key, without copying the key to any EC2 instance.

• Validate ICMP connectivity between App-A and App-B and, if required by the project rubric, perform the controlled ICMP security-group failure and restoration test.

• Insert the final evidence screenshots into the project folder and update this report if additional tests change the results.

• Publish the project to GitHub without the PEM key, credentials, or sensitive account information.

• Terminate the lab EC2 instances and remove lab-only networking resources after evidence and report are safely saved, to avoid ongoing AWS charges.

6. Conclusion

The two VPCs, subnets, route tables, security groups, EC2 instances, and VPC peering connection were deployed. Browser-based access to the bastion was established, and TCP port 22 from the bastion to both private instances was verified. ICMP replies and laptop-to-bastion SSH remained unresolved at the time this report was prepared; therefore, full end-to-end SSH and ping success should not be claimed until those tests pass.

Evidence folder: screenshots/ — add the corresponding implementation and test screenshots before final submission.